# Notes: Geng & Pan (JF, 2024)

## The SOE Premium and Government Support in China's Credit Market

# Contents

# 1 Big picture

- **Question.** How large is the funding advantage of state-owned enterprises (SOEs) in China's credit market, and why did this advantage explode after the 2018 asset-management regulations?
- **Core object.** The paper measures the **SOE premium**: the difference between non-SOE and SOE corporate bond credit spreads after controlling for ratings and issuer/bond characteristics.
- **Main empirical fact.** Before 2018, the listed-firm SOE premium is moderate, around 20–30 basis points. After the April 2018 New Regulations, it jumps sharply, reaching about 98 bps in 2018Q2 and peaking around 154 bps in 2019Q3.
- **Mechanism.** The regulations shrink shadow-banking and asset-management channels, creating a credit-market liquidity shock. Government support becomes more valuable in a liquidity crisis because it mitigates liquidity-driven default for SOEs.
- **Model contribution.** The paper extends the He and Xiong rollover-risk model by adding bailout probability. The interaction between liquidity risk and bailout probability generates a large SOE premium even if the strength of government support itself is stable.
- **Default measurement.** The authors construct a **unified default measure** that integrates fundamental credit risk, liquidity, and government support. It explains the SOE premium better than a standard Merton default measure.
- **Price discovery result.** In Phase III, credit spreads become much more sensitive to default measures. SOEs become more sensitive to government support; non-SOEs become more sensitive to credit quality.
- **Real effect.** Non-SOE operating performance deteriorates sharply relative to SOEs after 2018Q2. This reverses the long-standing pattern in which non-SOEs outperform SOEs.
- **Economic takeaway.** The same liquidity shock can widen credit misallocation when some firms have bailout protection and others do not. In China, the value of government support rises precisely when market liquidity deteriorates.

# 2 Data and measurement

## 2.1 Credit-market data

- The main bond-level sample runs from January 2010 to June 2020.
- The paper studies bonds issued by Chinese listed firms and traded in China's credit market.
- It focuses on credit spreads, ratings, bond maturity, issue size, callability, puttability, exchange-traded status, zero-trading days, and issuer equity information.
- SOE status separates listed issuers into SOEs and non-SOEs.
- The sample is split into three phases:
  - Phase I: 2010–2013, before defaults become salient;
  - Phase II: 2014–2018Q1, the first wave of defaults;
  - Phase III: 2018Q2–2020Q2, after the April 2018 New Regulations.

**Interpretation:** The phase split is central. The paper is not only asking whether SOEs have lower spreads, but whether the value of SOE status changes when the credit market becomes less liquid.

## 2.2 The SOE premium

The empirical SOE premium is measured from quarterly regressions of credit spreads on the non-SOE indicator and controls:

$$Spread_{i,t} = \alpha + \beta\, NSOE_{i,t} + \Gamma' X_{i,t} + \text{quarter FE} + \text{industry FE} + \varepsilon_{i,t}. \tag{1}$$

- $Spread_{i,t}$ is the credit spread on bond $i$ in quarter $t$.
- $NSOE_{i,t} = 1$ for non-SOE issuers and zero for SOE issuers.
- $X_{i,t}$ includes bond and issuer controls such as rating, maturity, issue size, liquidity proxies, and equity size.
- The coefficient $\beta$ is the SOE premium: non-SOE spreads relative to SOE spreads, conditional on observables.

**Interpretation:** A positive coefficient on $NSOE$ means investors require higher yields from non-SOEs than from otherwise similar SOEs. The premium is interpreted as the bond-market value of perceived government support.

## 2.3 The 2018 New Regulations

- In April 2018, China introduced new asset-management regulations aimed at reducing shadow-banking risks.
- The rules imposed net asset valuation, restricted guarantees, reduced maturity mismatch, and limited nested/fund-pooling structures.
- Although aimed at the asset-management sector, the rules transmitted quickly into the corporate bond market because asset managers were important holders and intermediaries.
- The paper documents liquidity deterioration, reduced bond-market access, and rising defaults after the reform, especially for non-SOEs.

**Interpretation:** The regulations are not simply a policy dummy. They create a market-wide liquidity shock that reveals differences in bailout protection.

## 2.4 Government support and government holdings

- For SOEs, government support is proxied by government ownership and holdings.
- The model interprets this as a bailout probability $\pi_g$.
- Non-SOEs are treated as having zero bailout probability in the benchmark model.
- SOEs differ in the magnitude of government support, allowing the paper to ask whether higher support matters more after the liquidity shock.

**Interpretation:** The key heterogeneity is not only SOE versus non-SOE. Among SOEs, greater government support should become more valuable when liquidity risk increases.

## 2.5 Default measures

The paper compares two measures of default risk:

- **Merton default measure:** captures fundamental credit quality using firm value and asset volatility.
- **Unified default measure:** integrates fundamental credit quality, liquidity risk, and bailout probability.

**Interpretation:** The standard Merton measure can identify weak fundamentals, but it cannot capture the way liquidity risk and bailout expectations jointly change default risk in China's credit market.

### 2.6 Real outcomes

- The paper studies quarterly ROA and ROE for listed SOEs and non-SOEs.
- It examines whether credit deterioration after 2018Q2 predicts subsequent operating performance.
- It also sorts firms by changes in default measures to connect credit-market signals to real effects.

**Interpretation:** If the bond-market SOE premium reflects real credit allocation rather than only pricing noise, it should predict changes in firm performance.

## 3 Models and empirical specifications

### 3.1 Firm asset process

The firm value follows a geometric Brownian motion under the risk-neutral measure:

$$\frac{dV_t}{V_t} = (r - \delta)dt + \sigma dZ_t, \tag{2}$$

where $r$ is the risk-free rate, $\delta$ is the payout rate, $\sigma$ is asset volatility, and $Z_t$ is Brownian motion.

**Interpretation:** This is the standard structural-credit starting point. The innovation is not the asset process but the combination of rollover liquidity risk and government bailout.

### 3.2 Debt structure and liquidity shock

The firm commits to a stationary debt structure $(C, P, m)$:

- $C$ is total annual coupon payment;
- $P$ is total principal;
- $m$ is the maturity of newly issued debt;
- the firm continuously issues new bonds to roll over maturing bonds.

The bond price satisfies a PDE with a liquidity-cost term:

$$rd(V_t,\tau) = c - \xi k d(V_t,\tau) - \frac{\partial d(V_t,\tau)}{\partial \tau} + (r-\delta)V_t\frac{\partial d(V_t,\tau)}{\partial V} + \frac{1}{2}\sigma^2 V_t^2 \frac{\partial^2 d(V_t,\tau)}{\partial V^2}. \tag{3}$$

**Interpretation:** The term $\xi k d(V_t,\tau)$ is the liquidity loss. Higher $\xi$ worsens the cost of rolling over debt and raises default risk.

### 3.3 Government bailout boundary condition

Without bailout, a defaulted bond recovers $\alpha V_B/m$. With government support, the boundary condition becomes

$$d(V_B,\tau,\pi_g;V_B) = \frac{\alpha V_B}{m}(1-\pi_g) + \pi_g\frac{P}{m}. \tag{4}$$

**Interpretation:** With probability $\pi_g$, the government helps bondholders receive full principal. Higher $\pi_g$ raises bond value and lowers credit spreads.

## 3.4 Equity valuation and endogenous default

The default boundary is chosen endogenously by equityholders. Equity value satisfies

$$rE = (r-\delta)V_t E_V + \frac{1}{2}\sigma^2 V_t^2 E_{VV} + \delta V_t - (1-\tau)C + \big[d(V_t, m, \pi_g) - p\big]. \tag{5}$$

Default occurs when

$$E(V_B) = 0. \tag{6}$$

**Interpretation:** Liquidity risk enters through the rollover gain/loss $d(V_t, m, \pi_g) - p$. If new debt is issued at a discount, equityholders bear a rollover loss and default earlier.

## 3.5 Core mechanism of the model

The model's central implication is

$$\text{higher liquidity shock } \xi \quad\Rightarrow\quad \text{higher liquidity-driven default risk} \quad\Rightarrow\quad \text{larger value of bailout } \pi_g. \tag{7}$$

**Interpretation:** The model does not need government support to become stronger after 2018. It needs only the same support to become more valuable when liquidity risk becomes severe.

## 3.6 Explaining the SOE premium with default measures

The paper augments the SOE premium regression with default measures:

$$Spread_{i,t} = \alpha + \beta NSOE_{i,t} + \theta DM_{i,t} + \Gamma' X_{i,t} + FE_t + FE_j + \varepsilon_{i,t}. \tag{8}$$

- Using Merton's $DM$ leaves a large Phase III SOE premium.
- Using the unified $DM$ greatly reduces the unexplained SOE premium.

**Interpretation:** The unified default measure internalizes the mechanism: government support and liquidity risk change the effective default risk of otherwise similar firms.

## 3.7 Price-discovery regressions

The paper estimates how credit spreads load on default measures and government support:

$$Spread_{i,t} = \alpha + \theta_1 MertonDM_{i,t} + \theta_2 GovtHoldings_{i,t} + \theta_3 UnifiedDM_{i,t} + \Gamma' X_{i,t} + FE + \varepsilon_{i,t}. \tag{9}$$

**Interpretation:** If the credit market becomes more informative, spreads should load more strongly on relevant risk variables. The paper finds this pattern after 2018Q2.

## 3.8 Real-performance regressions

For real effects, the paper estimates quarterly performance regressions such as

$$ROA_{i,t} = \alpha + \beta NSOE_{i,t} + \gamma GovtHoldings_{i,t} + \lambda EquitySize_{i,t} + FE_t + FE_j + \varepsilon_{i,t}. \tag{10}$$

For post-event changes, it uses

$$\Delta ROA_{i,t+\tau} = ROA_{i,t+\tau} - \overline{ROA}_{i,t-1}, \tag{11}$$

where $\overline{ROA}_{i,t-1}$ averages preevent ROA over prior quarters.

**Interpretation:** This connects bond-market deterioration to actual operating performance after the 2018Q2 shock.

# 4 Identification and empirical design

## 4.1 What identifies the main pricing facts

- The SOE premium is identified by comparing credit spreads of SOE and non-SOE bonds with similar ratings and characteristics within the same quarter and industry.
- The sharp change after 2018Q2 uses the timing of the New Regulations as a credit-market liquidity shock.
- The comparison across phases shows whether SOE status becomes more valuable when liquidity risk increases.

**Interpretation:** The design is not a randomized experiment, but the post-2018Q2 shock provides a powerful timing event that separates normal conditions from a liquidity-stress regime.

## 4.2 Why ratings alone are not enough

- The regressions control for ratings, but the SOE premium remains large.
- The post-2018Q2 spread gap reflects differences not captured by ratings.
- The unified default measure shows that liquidity risk plus bailout expectations can explain much of this gap.

**Interpretation:** Ratings summarize credit quality imperfectly in a market where implicit government support is central.

## 4.3 Why the unified default measure matters

- Merton's default measure focuses on fundamentals.
- The unified default measure combines fundamentals with bailout and liquidity-driven default.
- It explains the Phase III premium much better and makes spreads load more symmetrically on default risk.

**Interpretation:** The model is not only a theoretical story. Its default measure has empirical explanatory power that the standard model lacks.

## 4.4 Real-effects identification

- The paper examines performance changes after 2018Q2, controlling for preevent performance and firm characteristics.
- It compares non-SOEs and SOEs, and sorts firms by credit deterioration.
- It asks whether the market-implied credit shock predicts subsequent ROA/ROE declines.

**Interpretation:** The real-effects evidence supports the interpretation that the widening SOE premium reflects real credit conditions, not only transitory pricing noise.

### 4.5 Limits

- SOE status and government ownership are not randomly assigned.
- Bond-market pricing may reflect investor beliefs about support, not actual bailout probabilities.
- The credit market is only one part of China's broader credit system, though it is informative because prices are observable.
- The model imposes structure to separate credit, liquidity, and bailout channels.

**Interpretation:** The evidence is strongest as a coherent combination of facts: timing, model mechanism, price discovery, and real performance all point to the same conclusion.

## 5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped to save space and speed up comparison. All visuals are directly cropped from the paper.

### 5.1 Table I and Table II: sample characteristics and the baseline SOE premium

**Read: Table I.**

- The sample covers listed SOE and non-SOE bonds across three phases.
- Non-SOE credit spreads rise dramatically from Phase II to Phase III; SOE spreads rise much less.
- Turnover falls more for non-SOE bonds, consistent with stronger liquidity deterioration.
- The equity-level panel reports differences in size, leverage, growth, volatility, and default measures across issuer types.

**Read: Table II.**

- In the listed sample, the non-SOE coefficient is 0.20 in Phase I, 0.27 in Phase II, and 1.13 in Phase III.
- In the unlisted sample, the non-SOE coefficient rises from 0.25 in Phase I to 0.91 in Phase II and 1.81 in Phase III.
- The widening premium survives rating, maturity, liquidity, and bond-structure controls.

**Economic message:** The basic pricing fact is sharp: after 2018Q2, non-SOE bonds trade at much higher spreads than otherwise comparable SOE bonds.

**Table I**

**Summary Statistics: Sample of Corporate Bonds Used for Empirical Tests on Credit Pricing**

The sample period is from January 2010 to June 2020. Phase I, from 2010 through 2013, is the predefault period. Phase II, from 2014 through 2018Q1, captures the first wave of defaults. Phase III, from 2018Q2 to 2020Q2, captures the second and more severe wave of defaults. *CreditSpread* is the difference in yield between corporate bonds and Chinese Development Bank bonds of the same maturity. *Rating* is a numerical number: 1 = AAA, 2 = AA+, 3 = AA, 4 = AA-, etc. *Exch* is equal to one for exchange-traded bonds. *Callable* is equal to one for bonds issued with callable options. *Puttable* is equal to one for bonds issued with puttable options. *ZeroDays* is the percent of nontrading days per quarter. *Turnover* is the ratio of quarterly trading volume to issuance size. *TradingDays* is the number of trading days per quarter. *EquitySize* is the log of the firm's equity size. *Leverage* is the ratio of total current liabilities plus total noncurrent liabilities to total asset value. *AssetGrowth* is the average growth rate of the asset value in the past three years. *EquityVolatility* is the annualized equity volatility. *GovtHoldings* is the total government equity holdings within the firm's top 10 shareholders. *Merton DM* is the inverse of Merton's distance-to-default. *Unified DM* is the inverse of the distance-to-default from our unified model. *AssetValue* and *AssetVolatility* are model-implied asset value and annualized asset volatility, respectively.

Panel A: Bond-Level Variables

| | All | | | | Phase I | | | | Phase II | | | | Phase III | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | NSOE | | SOE | | NSOE | | SOE | | NSOE | | SOE | | NSOE | | SOE | |
| | mean | std | mean | std | mean | std | mean | std | mean | std | mean | std | mean | std | mean | std |
| NumBonds | 915 | | 1471 | | 218 | | 452 | | 636 | | 819 | | 564 | | 884 | |
| *CreditSpread* (%) | 2.46 | 2.38 | 1.39 | 1.41 | 2.03 | 1.25 | 1.21 | 0.79 | 2.04 | 1.40 | 1.32 | 1.31 | 3.54 | 3.74 | 1.70 | 1.89 |
| *Rating* | 2.43 | 0.85 | 1.68 | 0.84 | 2.73 | 0.75 | 1.83 | 0.86 | 2.60 | 0.73 | 1.79 | 0.88 | 1.91 | 0.91 | 1.34 | 0.61 |
| *Maturity* (yr) | 2.96 | 1.24 | 3.33 | 1.70 | 3.89 | 1.37 | 4.17 | 2.02 | 2.92 | 1.16 | 3.22 | 1.56 | 2.43 | 0.94 | 2.76 | 1.31 |
| *IssueSize* (billions) | 1.03 | 0.89 | 2.01 | 2.57 | 0.94 | 0.80 | 2.34 | 3.18 | 1.00 | 0.93 | 1.90 | 2.49 | 1.14 | 0.85 | 1.91 | 2.01 |
| *Age* (yr) | 1.75 | 1.26 | 2.01 | 1.67 | 1.25 | 1.04 | 1.54 | 1.37 | 1.81 | 1.31 | 2.26 | 1.69 | 1.94 | 1.21 | 1.98 | 1.77 |
| *Coupon* (%) | 5.90 | 1.24 | 5.12 | 1.09 | 6.45 | 0.99 | 5.43 | 0.98 | 5.94 | 1.23 | 5.23 | 1.09 | 5.46 | 1.25 | 4.65 | 1.05 |
| *Exch* | 0.69 | 0.46 | 0.53 | 0.50 | 0.77 | 0.42 | 0.56 | 0.50 | 0.69 | 0.46 | 0.56 | 0.50 | 0.63 | 0.48 | 0.45 | 0.50 |
| *Callable* | 0.04 | 0.20 | 0.08 | 0.27 | 0.00 | 0.00 | 0.02 | 0.14 | 0.04 | 0.21 | 0.08 | 0.27 | 0.06 | 0.24 | 0.14 | 0.34 |
| *Puttable* | 0.62 | 0.49 | 0.34 | 0.47 | 0.55 | 0.50 | 0.29 | 0.45 | 0.64 | 0.48 | 0.37 | 0.48 | 0.63 | 0.48 | 0.31 | 0.46 |
| *ZeroDays* (%) | 77.31 | 26.06 | 85.75 | 18.47 | 62.30 | 30.00 | 79.00 | 21.44 | 76.71 | 26.20 | 85.44 | 19.29 | 88.32 | 15.82 | 92.36 | 9.70 |
| *Turnover* (%) | 30.69 | 62.29 | 35.39 | 80.68 | 44.38 | 91.30 | 54.43 | 118.52 | 31.44 | 56.11 | 30.81 | 70.16 | 20.25 | 46.65 | 26.50 | 46.33 |
| *TradingDays* (day) | 15 | 17 | 10 | 12 | 25 | 20 | 14 | 14 | 16 | 18 | 10 | 13 | 8 | 11 | 5 | 6 |

(*Continued*)

Table I, Panel A: bond-level summary statistics.

**Table I**—*Continued*

Panel B: Equity-Level Variables

| | All | | | | Phase I | | | | Phase II | | | | Phase III | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NumIssuers | 365 | | 401 | | 176 | | 251 | | 314 | | 338 | | 225 | | 252 | |
| *EquitySize* (log) | 23.31 | 1.03 | 23.72 | 1.33 | 22.57 | 0.93 | 23.33 | 1.40 | 23.31 | 0.88 | 23.72 | 1.28 | 23.78 | 1.08 | 24.05 | 1.28 |
| *Leverage* (%) | 58.50 | 15.30 | 61.75 | 14.87 | 55.75 | 12.85 | 61.26 | 14.63 | 57.32 | 15.29 | 61.30 | 15.63 | 62.65 | 15.98 | 63.00 | 13.56 |
| *AssetGrowth* (%) | 24.91 | 19.40 | 14.33 | 13.04 | 28.55 | 21.06 | 19.76 | 14.27 | 24.60 | 19.69 | 12.85 | 12.99 | 23.14 | 17.27 | 12.11 | 10.37 |
| *EquityVolatility* (%) | 40.39 | 17.95 | 36.34 | 18.37 | 37.55 | 10.05 | 32.41 | 10.80 | 42.34 | 21.46 | 41.55 | 22.35 | 38.41 | 13.33 | 30.51 | 12.17 |
| *GovtHoldings* (%) | 5.09 | 8.40 | 52.00 | 16.74 | 4.98 | 8.74 | 52.29 | 17.32 | 4.54 | 7.55 | 51.26 | 16.68 | 6.24 | 9.57 | 53.08 | 16.26 |
| *Merton DM* (%) | 21.19 | 12.79 | 22.56 | 15.16 | 18.72 | 6.59 | 18.33 | 7.82 | 22.55 | 15.02 | 26.81 | 18.95 | 20.14 | 10.55 | 18.71 | 9.26 |
| *AssetValue* (log) | 24.01 | 1.24 | 24.66 | 1.46 | 23.19 | 0.97 | 24.27 | 1.45 | 23.90 | 1.04 | 24.54 | 1.40 | 24.77 | 1.31 | 25.23 | 1.40 |
| *AssetVolatility* (%) | 22.99 | 15.49 | 17.12 | 13.81 | 22.15 | 10.35 | 14.97 | 9.48 | 26.14 | 17.43 | 21.34 | 16.23 | 17.32 | 12.23 | 11.46 | 9.07 |
| *Unified DM* (%) | 33.43 | 13.46 | 21.94 | 9.15 | 31.71 | 11.12 | 21.21 | 8.06 | 32.04 | 13.42 | 22.96 | 10.34 | 37.31 | 14.17 | 20.77 | 7.45 |
| *AssetValue* (log) | 24.16 | 1.27 | 24.78 | 1.51 | 23.30 | 0.97 | 24.36 | 1.48 | 24.01 | 1.05 | 24.58 | 1.42 | 25.01 | 1.33 | 25.50 | 1.45 |
| *AssetVolatility* (%) | 17.50 | 14.81 | 16.67 | 15.54 | 16.44 | 10.38 | 14.07 | 10.11 | 20.64 | 16.73 | 21.90 | 18.48 | 12.01 | 11.08 | 9.60 | 9.06 |

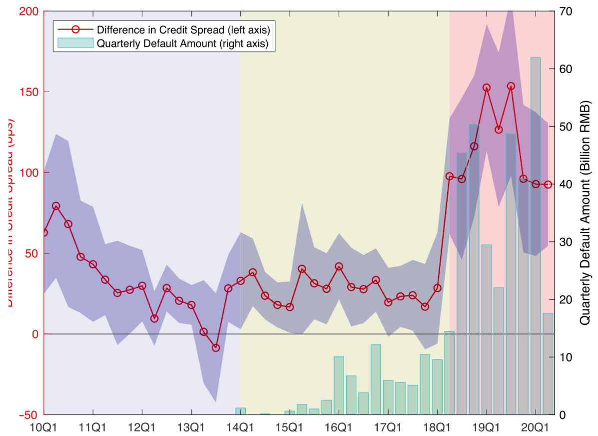
Table I, Panel B: issuer-level summary statistics.

**Table II**

**Measuring the SOE Premium**

This table reports the result of quarterly panel regressions of credit spreads on non-SOE dummy (*NSOE*) and other controls with quarter and industry fixed effects. *NSOE* is equal to one for bonds issued by non-SOEs and zero for SOEs. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. See Table I for bond-level variable definitions. The sample period is from January 2010 to June 2020. Phase I, from 2010 through 2013, is the predefault period. Phase II, from 2014 through 2018Q1, captures the first wave of defaults. Phase III, from 2018Q2 to 2020Q2, captures the second and more severe wave of defaults. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

| | Listed Sample | | | Unlisted Sample | | |
|---|---|---|---|---|---|---|
| | Phase I | Phase II | Phase III | Phase I | Phase II | Phase III |
| *NSOE* | 0.20*** | 0.27*** | 1.13*** | 0.25*** | 0.91*** | 1.81*** |
| | [2.97] | [4.28] | [7.76] | [5.65] | [15.26] | [17.87] |
| *Rating* | 0.52*** | 0.53*** | 1.19*** | 0.49*** | 0.47*** | 0.48*** |
| | [6.45] | [10.62] | [5.12] | [14.86] | [17.52] | [14.82] |
| *Maturity* | 0.05*** | 0.05** | −0.08 | 0.03*** | −0.02 | −0.16*** |
| | [2.82] | [2.43] | [−1.41] | [2.97] | [−1.41] | [−12.21] |
| *Age* | 0.03 | 0.04** | 0.04 | 0.01 | 0.00 | 0.05*** |
| | [1.30] | [2.02] | [0.95] | [0.68] | [0.23] | [3.24] |
| *IssueSize* | 0.00 | −0.04*** | −0.08** | −0.03*** | −0.08*** | −0.18*** |
| | [0.01] | [−2.82] | [−2.09] | [−6.93] | [−9.45] | [−10.15] |
| *ZeroDays* | −0.92*** | −2.10*** | −4.34*** | −0.43*** | −1.70*** | −3.07*** |
| | [−3.92] | [−9.35] | [−5.84] | [−3.81] | [−8.55] | [−8.44] |
| *Exch* | 0.03 | −0.46*** | −0.03 | −0.09* | −0.36*** | −0.07 |
| | [0.39] | [−6.59] | [−0.14] | [−1.85] | [−6.04] | [−0.72] |
| *Callable* | −0.24** | 1.55*** | 2.25*** | 0.17* | 2.21*** | 2.65*** |
| | [−2.00] | [8.17] | [9.08] | [1.78] | [18.22] | [25.55] |
| *Puttable* | −0.16** | −0.28*** | −0.72*** | 0.33*** | 0.12** | 0.15** |
| | [−2.12] | [−3.34] | [−3.72] | [5.04] | [2.53] | [2.19] |
| *EquitySize* | −0.15*** | −0.19*** | −0.26*** | | | |
| | [−5.15] | [−5.24] | [−4.49] | | | |
| Constant | 4.26*** | 6.27*** | 9.72*** | 0.93*** | 1.54*** | 3.23*** |
| | [4.42] | [7.13] | [5.29] | [5.57] | [6.61] | [8.06] |
| Obs | 4,292 | 9,967 | 5,338 | 16,179 | 32,240 | 15,833 |
| Adj $R^2$ | 0.546 | 0.455 | 0.376 | 0.561 | 0.508 | 0.491 |




**ure 1. The SOE premium**. This figure plots the difference between listed non-SOEs and ed SOEs in credit spread (left axis), estimated using quarterly regressions, controlling for credit ings and other bond and firm characteristics. The shaded area indicates 95% confidence inter- s. Also reported are the total quarterly default amounts in the credit market (right axis). (Color ure can be viewed at wileyonlinelibrary.com)

## 5.2 Figure 1: the SOE premium over time

**Read:**

- Before 2018, the SOE premium fluctuates at a moderate level.
- It jumps sharply in 2018Q2 after the New Regulations.
- Defaults rise at the same time, especially in the later Phase III period.
- The premium remains high even after policy reassurances to support private firms.

**Economic message:** The SOE premium is time-varying and crisis-sensitive. Government support becomes most valuable when liquidity and default risk intensify.

## 5.3 Figure 2 and Table III: model calibration

**Read:**

- Credit spreads fall with bailout probability $\pi_g$ and rise with liquidity-shock intensity $\xi$.
- The relation is nonlinear: bailout protection matters much more when liquidity risk is high.
- When $\xi$ rises from 1 to 2, the premium for a non-SOE versus an SOE with $\pi_g = 0.6$ almost doubles.
- Panel C of Table III shows that default probabilities rise more for non-SOEs than for SOEs when liquidity risk increases.

**Economic message:** The model can generate an explosive SOE premium without assuming that government support itself changes. Liquidity stress increases the market value of the existing bailout option.

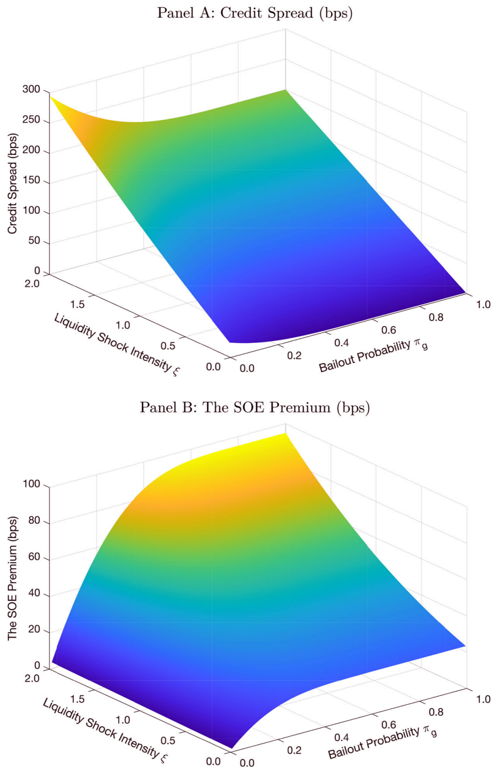



Figure 2: Model calibrations on the SOE premium.

**Table III**

## Calibration Results

This table reports the credit spread (Panel A), SOE premium (Panel B), and one-year default probability (Panel C) with varying bailout probability $\pi_g$ and liquidity shock $\xi$. Panel D reports the SOE premium with varying liquidity shock $\xi$ and maturity $m$ when $\pi_g^{SOE} = 0.6$. The SOE premium is defined as the difference between non-SOEs ($\pi_g = 0$) and SOEs with positive $\pi_g$. We set the risk-free rate $r$ to 4%, the tax rate $\tau$ to 25%, the bond recovery rate $\alpha$ to 50%, the payout rate $\Delta$ to 2%, asset volatility $\sigma$ to 15%, asset growth $\mu$ to 15%, the maturity of the bond $m$ to one year, the trading cost $k$ to 1%, and the current fundamental value of the firm to 100.

Panel A: Credit Spread with Varying Bailout Probability $\pi_g$ and Liquidity Shock $\xi$

| | $\xi$=1 | $\xi$ Rises to 2 | | | | $\xi$ Rises to 3 | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Spread | Spread | △Spread | Default Part | | Spread | △Spread | Default Part | |
| | (bps) | (bps) | (bps) | (bps) | (fraction) | (bps) | (bps) | (bps) | (fraction) |
| NSOE: $\pi_g = 0$ | 150 | 295 | 145 | 45 | 31.0% | 459 | 309 | 109 | 35.2% |
| SOE: $\pi_g = 0.2$ | 116 | 239 | 123 | 23 | 18.6% | 379 | 263 | 63 | 23.9% |
| SOE: $\pi_g = 0.4$ | 103 | 210 | 107 | 7 | 6.7% | 328 | 225 | 25 | 11.1% |
| SOE: $\pi_g = 0.6$ | 100 | 201 | 101 | 1 | 0.9% | 305 | 205 | 5 | 2.4% |
| SOE: $\pi_g = 0.8$ | 100 | 200 | 100 | 0 | 0.0% | 300 | 200 | 0 | 0.1% |
| SOE: $\pi_g = 1.0$ | 100 | 200 | 100 | 0 | 0.0% | 300 | 200 | 0 | 0.0% |

Panel B: The SOE Premium with Varying Bailout Probability $\pi_g$ and Liquidity Shock $\xi$

| | $\xi$=1 | $\xi$ Rises to 2 | | | $\xi$ Rises to 3 | | |
|---|---|---|---|---|---|---|---|
| | Premium | Premium | △Premium | | Premium | △Premium | |
| NSOE − SOE | (bps) | (bps) | (bps) | (multiplier) | (bps) | (bps) | (multiplier) |
| $\pi_g = 0 - \pi_g = 0.2$ | 34 | 56 | 22 | 1.7 | 80 | 46 | 2.3 |
| $\pi_g = 0 - \pi_g = 0.4$ | 47 | 85 | 38 | 1.8 | 131 | 84 | 2.8 |
| $\pi_g = 0 - \pi_g = 0.6$ | 50 | 94 | 44 | 1.9 | 154 | 104 | 3.1 |
| $\pi_g = 0 - \pi_g = 0.8$ | 50 | 95 | 45 | 1.9 | 159 | 109 | 3.2 |
| $\pi_g = 0 - \pi_g = 1.0$ | 50 | 95 | 45 | 1.9 | 159 | 109 | 3.2 |

Panel C: One-Year Default Probability with Varying Bailout Probability $\pi_g$ and Liquidity Shock $\xi$

| | $\xi$=1 | $\xi$ Rises to 2 | | $\xi$ Rises to 3 | |
|---|---|---|---|---|---|
| | Prob (bps) | Prob (bps) | △Prob (bps) | Prob (bps) | △Prob (bps) |
| NSOE: $\pi_g = 0$ | 25 | 67 | 42 | 160 | 135 |
| SOE: $\pi_g = 0.2$ | 7 | 24 | 17 | 73 | 66 |
| SOE: $\pi_g = 0.4$ | 1 | 5 | 4 | 23 | 22 |
| SOE: $\pi_g = 0.6$ | 0 | 0 | 0 | 4 | 4 |
| SOE: $\pi_g = 0.8$ | 0 | 0 | 0 | 0 | 0 |
| SOE: $\pi_g = 1.0$ | 0 | 0 | 0 | 0 | 0 |

*(Continued)*

Table III: Calibration results, panels A–C.

**Table III**—*Continued*

Panel D: The SOE Premium with Varying Liquidity Shock $\xi$ and Maturity $m$ When $\pi_g^{SOE} = 0.6$

| | $\xi$=1 | $\xi$ Rises to 2 | | | $\xi$ Rises to 3 | | |
|---|---|---|---|---|---|---|---|
| | Premium | Premium | △Premium | | Premium | △Premium | |
| NSOE − SOE | (bps) | (bps) | (bps) | (multiplier) | (bps) | (bps) | (multiplier) |
| $m = 1$ | 50 | 94 | 44 | 1.9 | 154 | 104 | 3.1 |
| $m = 3$ | 48 | 74 | 26 | 1.5 | 100 | 52 | 2.1 |
| $m = 6$ | 45 | 59 | 15 | 1.3 | 73 | 28 | 1.6 |
| $m = 10$ | 41 | 50 | 9 | 1.2 | 58 | 17 | 1.4 |

At each quarter $t$ and for each issuer $i$, the empirical construction of $DM_{t,i}$ feeds the model firm $i$'s balance sheet and equity information in quarter $t$, and then brings back the model-implied DM. We start our exposition with Merton's DM, and then move on to construct the DM implied by our model, incorporating the issuer-level government support (i.e., $\pi_g$) and the economy-wide liquidity condition (i.e., $\xi$). We refer to it as a unified DM, as it combines information on credit, liquidity, and government support into a single measure.

*A.1. Merton's DM*

Within the structural model of Merton (1974), the one-year DD under the actual probability measure is

$$DD_t^{Merton} = \frac{\ln(V_t/K) + \left(\mu - \sigma_A^2/2\right)}{\sigma_A} \quad \text{and} \quad DM_t^{Merton} = \left(DD_t^{Merton}\right)^{-1}, \qquad (10)$$

where $V_t$ is the time $t$ asset value of the firm, $\mu$ is its asset growth, $\sigma_A$ is its asset volatility, and $K$ is its default boundary. Under Merton's model, the default boundary is an exogenous concept, equal to the face value of the firm's debt outstanding. Although it is the risk-neutral DD that matters for credit price (i.e., replace $\mu$ by $r$), we follow convention by incorporating the issuer-level asset growth $\mu$ into the DM. Empirically, we find the asset growth to be informative in cross-sectional credit pricing.

Following Moody's KMV (Kealhofer and Kurbat (2001)), we estimate the

Table III continued: maturity and rollover-risk implications.

## 5.4 Figure 3, Table IV, and Figure 4: explaining the SOE premium

**Read: Figure 3.**

- Government holdings differ strongly across SOE and non-SOE types.
- Merton's default measure does not fully distinguish the post-2018 risk gap.
- The unified default measure produces a much clearer SOE/non-SOE separation.

**Read: Table IV and Figure 4.**

- Controlling for Merton's default measure leaves a large Phase III SOE premium.
- Controlling for the unified default measure substantially reduces the residual premium.
- Figure 4 shows this continuously over time: the blue line remains high post-2018, while the red line is much closer to zero.

**Economic message:** The unified default measure explains the premium because it embeds the economic mechanism: government support is valuable when liquidity-driven default is severe.

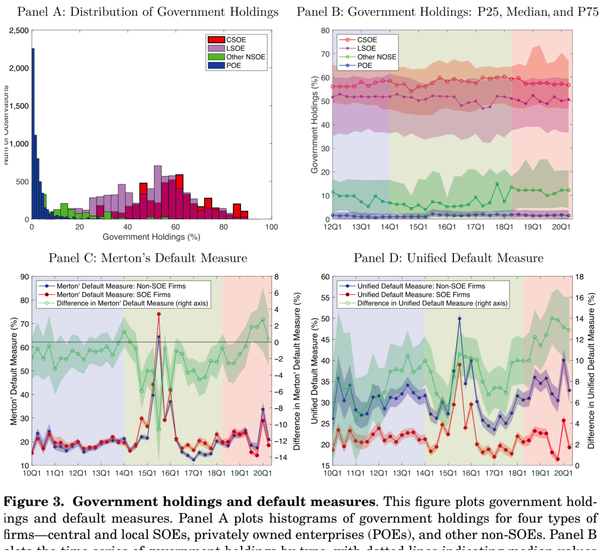



Figure 3: Government holdings and default measures.

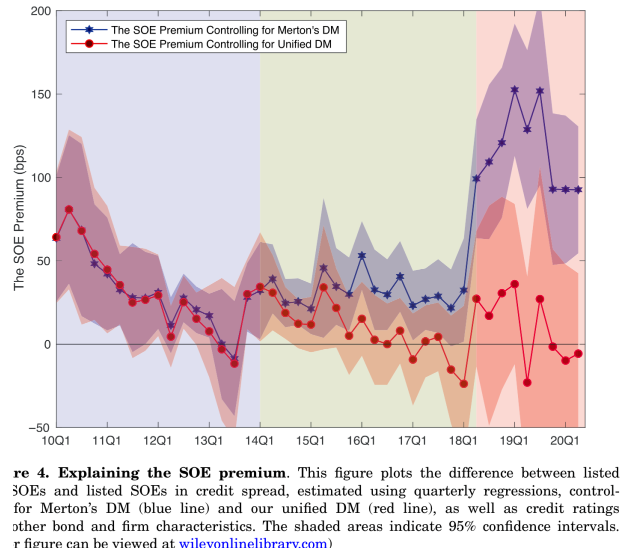



Figure 4: Explaining the SOE premium.

**Table IV**

**Explaining the SOE Premium**

This table reports the result of quarterly panel regressions of credit spreads on non-SOE dummy (*NSOE*), Merton's default measure (*Merton DM*), government holdings (*GovtHoldings*), or unified default measure (*Unified DM*), and other controls with quarter and industry fixed effects. *NSOE* is equal to one for bonds issued by non-SOEs and zero for SOEs. *Merton DM* is the inverse of Merton's distance-to-default. *GovtHoldings* is total government equity holdings within the firm's top 10 shareholders. *Unified DM* is the inverse of the distance-to-default from our unified model. Controls include bond maturity, issuance size, age, exchange market dummy, optionality, liquidity, and equity size. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. The sample period is from January 2010 to June 2020. Phase I, from 2010 through 2013, is the predefault period. Phase II, from 2014 through 2018Q1, captures the first wave of defaults. Phase III, from 2018Q2 to 2020Q2, captures the second and more severe wave of defaults. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

| | Phase I | | | | Phase II | | | | Phase III | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *NSOE* | 0.20*** | 0.20*** | 0.21** | 0.17** | 0.27*** | 0.32*** | 0.17 | 0.06 | 1.13*** | 1.16*** | −0.04 | 0.06 |
| | [2.97] | [2.84] | [2.57] | [1.96] | [4.28] | [5.05] | [1.51] | [0.82] | [7.76] | [7.88] | [−0.21] | [0.38] |
| *Merton DM* | | −0.12 | | | | 1.36*** | | | | 4.60*** | | |
| | | [−0.38] | | | | [4.52] | | | | [4.92] | | |
| *GovtHoldings* | | | 0.04 | | | | −0.26 | | | | −2.86*** | |
| | | | [0.23] | | | | [−1.24] | | | | [−6.95] | |
| *Unified DM* | | | | 0.34 | | | | 2.27*** | | | | 7.23*** |
| | | | | [0.91] | | | | [6.26] | | | | [9.92] |
| *Rating* | 0.52*** | 0.52*** | 0.52*** | 0.52*** | 0.53*** | 0.53*** | 0.52*** | 0.52*** | 1.19*** | 1.11*** | 1.15*** | 1.23*** |
| | [6.45] | [6.35] | [6.30] | [6.61] | [10.62] | [10.93] | [10.47] | [10.67] | [5.12] | [5.06] | [4.91] | [5.91] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Constant | 4.26*** | 4.30*** | 4.25*** | 3.99*** | 6.27*** | 5.42*** | 6.33*** | 4.93*** | 9.72*** | 7.42*** | 9.52*** | 3.59* |
| | [4.42] | [4.71] | [4.40] | [3.58] | [7.13] | [6.19] | [7.32] | [5.38] | [5.29] | [3.90] | [5.15] | [1.90] |
| Obs | 4,292 | 4,292 | 4,292 | 4,292 | 9,967 | 9,967 | 9,967 | 9,967 | 5,338 | 5,338 | 5,338 | 5,338 |
| Adj $R^2$ | 0.546 | 0.546 | 0.546 | 0.547 | 0.455 | 0.465 | 0.456 | 0.476 | 0.376 | 0.392 | 0.390 | 0.423 |

Table IV: Explaining the SOE premium.

## 5.5 Table V and Figure 5: price discovery

**Read:**

- Table V studies how spreads load on Merton default risk, government holdings, and the unified default measure.
- In Phase III, spreads become much more sensitive to default measures than in earlier phases.
- Figure 5 shows the slope coefficients from rolling quarterly regressions.
- The Merton measure generates a larger separation between SOEs and non-SOEs, while the unified measure produces a more coherent sensitivity pattern.

**Economic message:** China's credit market becomes more informative after the liquidity shock. Investors pay more attention to credit quality for non-SOEs and to government support for SOEs.

**Table V**

**Price Discovery: Credit Spreads on Default Measures and Government Holdings**

This table reports the result of quarterly panel regressions of credit spreads on Merton's default measure (*Merton DM*), government holdings (*GovtHoldings*), unified default measure (*Unified DM*), and other controls with quarter and industry fixed effects for non-SOEs sample (Panel A) and SOEs sample (Panel B). *Merton DM* is the inverse of Merton's distance-to-default. *GovtHoldings* is the total government equity holdings within the firm's top 10 shareholders. *Unified DM* is the inverse of the distance-to-default from our unified model. Controls include bond maturity, issuance size, age, exchange market dummy, optionality, liquidity, and equity size. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. The sample period is from January 2010 to June 2020. Phase I, from 2010 through 2013, is the predefault period. Phase II, from 2014 through 2018Q1, captures the first wave of defaults. Phase III, from 2018Q2 to 2020Q2, captures the second and more severe wave of defaults. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

Panel A: NSOE

| | Phase I | | | | | | Phase II | | | | | | Phase III | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Merton DM* | | −0.12 | | | −0.10 | −0.32 | | 1.59$^{***}$ | | | 1.60$^{***}$ | 0.87 | | 7.53$^{***}$ | | | 7.66$^{***}$ | 6.00$^{***}$ |
| | | [−0.14] | | | [−0.11] | [−0.32] | | [2.70] | | | [2.72] | [1.30] | | [3.67] | | | [3.82] | [3.64] |
| *GovtHoldings* | | | 0.52 | | 0.52 | 0.73 | | | −0.02 | | −0.13 | 0.26 | | | −6.00$^{***}$ | | −6.16$^{***}$ | −4.09$^{***}$ |
| | | | [1.14] | | [1.13] | [1.56] | | | [−0.05] | | [−0.28] | [0.60] | | | [−4.70] | | [−5.26] | [−3.48] |
| *Unified DM* | | | | 0.68 | | 0.85 | | | | 1.86$^{***}$ | | 1.58$^{***}$ | | | | 7.40$^{***}$ | | 4.90$^{***}$ |
| | | | | [1.03] | | [1.13] | | | | [4.61] | | [3.12] | | | | [5.88] | | [5.03] |
| *Rating* | 0.78$^{***}$ | 0.78$^{***}$ | 0.79$^{***}$ | 0.77$^{***}$ | 0.79$^{***}$ | 0.78$^{***}$ | 0.41$^{***}$ | 0.41$^{***}$ | 0.41$^{***}$ | 0.42$^{***}$ | 0.41$^{***}$ | 0.43$^{***}$ | 1.61$^{***}$ | 1.41$^{***}$ | 1.55$^{***}$ | 1.61$^{***}$ | 1.35$^{***}$ | 1.41$^{***}$ |
| | [3.12] | [3.12] | [3.20] | [3.17] | [3.19] | [3.26] | [4.44] | [4.56] | [4.49] | [4.60] | [4.58] | [4.72] | [4.35] | [4.13] | [4.27] | [4.93] | [3.96] | [4.47] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Constant | 7.55$^{***}$ | 7.57$^{***}$ | 7.53$^{***}$ | 6.69$^{***}$ | 7.55$^{***}$ | 6.50$^{***}$ | 11.60$^{***}$ | 10.77$^{***}$ | 11.60$^{***}$ | 9.83$^{***}$ | 10.76$^{***}$ | 9.67$^{***}$ | 15.19$^{***}$ | 13.21$^{***}$ | 11.44$^{***}$ | 4.44 | 9.34$^{***}$ | 3.95 |
| | [3.80] | [3.93] | [3.82] | [2.64] | [3.97] | [2.62] | [7.28] | [6.97] | [7.23] | [5.58] | [6.90] | [5.63] | [4.69] | [4.22] | [3.37] | [1.26] | [2.89] | [1.16] |
| Obs | 1,367 | 1,367 | 1,367 | 1,367 | 1,367 | 1,367 | 4,116 | 4,116 | 4,116 | 4,116 | 4,116 | 4,116 | 2,085 | 2,085 | 2,085 | 2,085 | 2,085 | 2,085 |
| Adj $R^2$ | 0.494 | 0.494 | 0.495 | 0.496 | 0.494 | 0.498 | 0.373 | 0.382 | 0.373 | 0.388 | 0.382 | 0.391 | 0.363 | 0.391 | 0.381 | 0.403 | 0.410 | 0.424 |

*(Continued)*

Panel B: SOE

| | Phase I | | | | | | Phase II | | | | | | Phase III | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Merton DM* | | 0.10 | | | 0.09 | 0.17 | | 1.21$^{***}$ | | | 1.21$^{***}$ | 0.47 | | 2.14$^{**}$ | | | 1.54$^{*}$ | −0.89 |
| | | [0.67] | | | [0.62] | [0.98] | | [3.95] | | | [3.95] | [1.54] | | [2.42] | | | [1.76] | [−1.17] |
| *GovtHoldings* | | | −0.15 | | −0.14 | −0.21 | | | −0.30 | | −0.31 | 0.20 | | | −2.23$^{***}$ | | −2.08$^{***}$ | −0.89$^{**}$ |
| | | | [−1.03] | | [−1.02] | [−1.48] | | | [−1.37] | | [−1.43] | [0.93] | | | [−5.41] | | [−5.27] | [−2.36] |
| *Unified DM* | | | | 0.04 | | −0.26 | | | | 2.37$^{***}$ | | 2.17$^{***}$ | | | | 6.86$^{***}$ | | 6.52$^{***}$ |
| | | | | [0.17] | | [−0.96] | | | | [4.26] | | [3.28] | | | | [6.66] | | [5.86] |
| *Rating* | 0.40$^{***}$ | 0.40$^{***}$ | 0.39$^{***}$ | 0.40$^{***}$ | 0.39$^{***}$ | 0.39$^{***}$ | 0.54$^{***}$ | 0.54$^{***}$ | 0.53$^{***}$ | 0.52$^{***}$ | 0.53$^{***}$ | 0.53$^{***}$ | 0.55$^{***}$ | 0.53$^{***}$ | 0.51$^{***}$ | 0.54$^{***}$ | 0.50$^{***}$ | 0.53$^{***}$ |
| | [11.25] | [11.22] | [11.13] | [11.36] | [11.11] | [11.06] | [9.01] | [9.37] | [8.97] | [9.37] | [9.30] | [9.41] | [4.52] | [4.41] | [4.48] | [4.81] | [4.40] | [4.80] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Constant | 3.33$^{***}$ | 3.29$^{***}$ | 3.38$^{***}$ | 3.30$^{***}$ | 3.35$^{***}$ | 3.53$^{***}$ | 4.78$^{***}$ | 3.95$^{***}$ | 4.81$^{***}$ | 3.31$^{***}$ | 3.98$^{***}$ | 3.10$^{***}$ | 8.16$^{***}$ | 6.80$^{***}$ | 8.49$^{***}$ | 3.33$^{**}$ | 7.49$^{***}$ | 4.27$^{***}$ |
| | [3.74] | [3.76] | [3.91] | [3.44] | [3.94] | [3.83] | [5.48] | [4.52] | [5.62] | [3.49] | [4.65] | [3.32] | [5.55] | [4.34] | [6.07] | [2.03] | [5.03] | [2.60] |
| Obs | 2,925 | 2,925 | 2,925 | 2,925 | 2,925 | 2,925 | 5,851 | 5,851 | 5,851 | 5,851 | 5,851 | 5,851 | 3,253 | 3,253 | 3,253 | 3,253 | 3,253 | 3,253 |
| Adj $R^2$ | 0.541 | 0.541 | 0.542 | 0.541 | 0.542 | 0.542 | 0.488 | 0.498 | 0.489 | 0.506 | 0.499 | 0.508 | 0.380 | 0.387 | 0.404 | 0.426 | 0.408 | 0.430 |

Table V, Panel A: non-SOEs.

Table V, Panel B: SOEs.

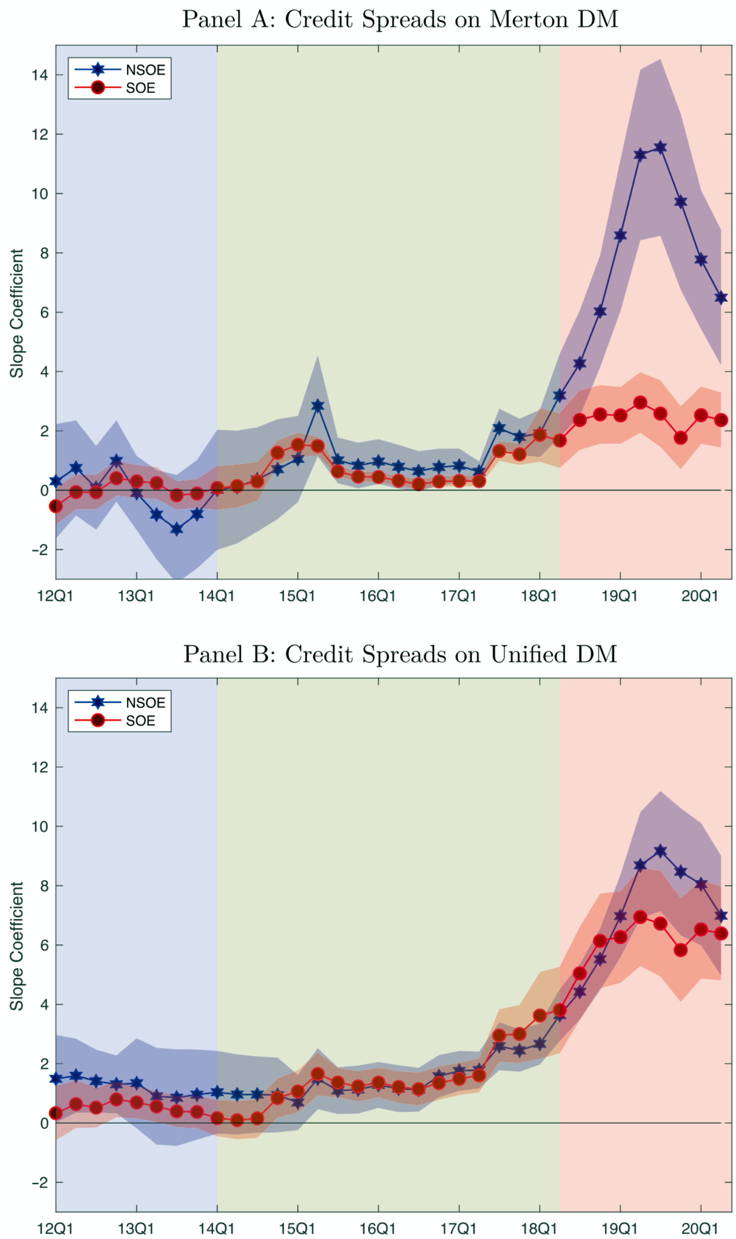



Figure 5: Slope coefficients from credit-spread regressions.

## 5.6 Figure 6 and Table VI: real performance of SOEs and non-SOEs

**Read:**

- Figure 6 shows that non-SOEs historically outperform SOEs in ROA.
- After 2018Q2, the non-SOE advantage collapses and in some quarters turns negative.
- Table VI shows this pattern in ROA and ROE regressions.
- Government holdings are negatively associated with performance in early phases, consistent with less efficient SOEs, but this relation weakens in Phase III.

**Economic message:** The credit shock has real consequences. The policy-induced liquidity tightening reverses the prior performance advantage of non-SOEs.

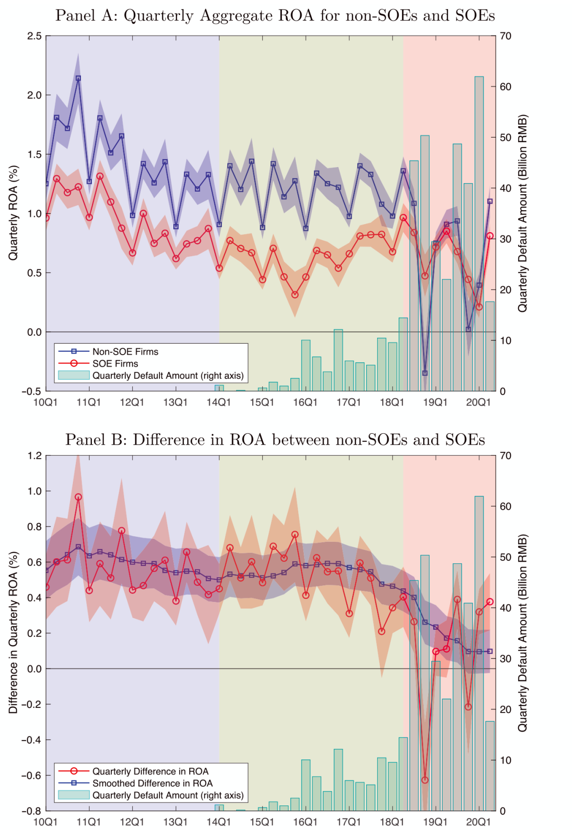



Figure 6: Quarterly ROA for non-SOEs and SOEs.

**Table VI**

**The Relative Performance between Non-SOEs and SOEs**

This table reports the result of quarterly panel regressions of ROA and ROE as the dependent variables on non-SOE dummy (*NSOE*), government holdings (*GovtHoldings*), and equity size with quarter and industry fixed effects. Panel B reports results within the matched sample using propensity score matching (PSM). *ROA* is the ratio of net profit to lagged book assets. *ROE* is net profit to lagged book equity. *NSOE* is equal to one for bonds issued by non-SOEs and zero for SOEs. *GovtHoldings* is total government equity holdings within the firm's top 10 shareholders. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. Phase I is from 2010 through 2013. Phase II is from 2014 through 2018Q1. Phase III is from 2018Q2 to 2020Q2. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

| | ROA (%) | | | ROE (%) | | |
|---|---|---|---|---|---|---|
| | Phase I | Phase II | Phase III | Phase I | Phase II | Phase III |
| **Panel A: NSOE Dummy** | | | | | | |
| *NSOE* | 0.56*** | 0.53*** | 0.13 | 1.08*** | 1.20*** | −0.01 |
| | [7.80] | [8.94] | [1.09] | [6.70] | [8.00] | [−0.03] |
| *EquitySize* | 0.19*** | 0.19*** | 0.35*** | 0.78*** | 0.74*** | 1.09*** |
| | [6.08] | [6.34] | [8.68] | [10.86] | [11.05] | [7.58] |
| Constant | −3.56*** | −4.05*** | −7.03*** | −16.25*** | −16.66*** | −21.40*** |
| | [−4.86] | [−4.95] | [−8.51] | [−9.75] | [−9.43] | [−7.94] |
| Obs | 15,677 | 18,487 | 10,844 | 15,677 | 18,487 | 10,844 |
| Adj $R^2$ | 0.065 | 0.064 | 0.095 | 0.051 | 0.046 | 0.085 |
| **Panel B: NSOE Dummy under PSM** | | | | | | |
| *NSOE* | 0.45*** | 0.38*** | −0.09 | 0.94*** | 0.96*** | −0.41 |
| | [4.06] | [4.24] | [−0.62] | [3.59] | [4.44] | [−0.87] |
| *EquitySize* | 0.19*** | 0.22*** | 0.42*** | 0.69*** | 0.62*** | 1.24*** |
| | [3.25] | [3.99] | [5.29] | [4.36] | [5.17] | [4.15] |
| Constant | −3.23** | −4.31*** | −8.41*** | −13.22*** | −12.76*** | −25.92*** |
| | [−2.55] | [−3.50] | [−4.72] | [−3.72] | [−4.67] | [−3.80] |
| Obs | 4,573 | 5,362 | 2,970 | 4,573 | 5,362 | 2,970 |
| Adj $R^2$ | 0.052 | 0.049 | 0.110 | 0.052 | 0.035 | 0.083 |
| **Panel C: Government Holdings** | | | | | | |
| *GovtHoldings* | −0.89*** | −0.91*** | −0.26 | −1.81*** | −2.09*** | 0.07 |
| | [−6.46] | [−7.93] | [−1.03] | [−5.77] | [−6.77] | [0.10] |
| *EquitySize* | 0.17*** | 0.21*** | 0.36*** | 0.76*** | 0.78*** | 1.09*** |
| | [5.74] | [6.84] | [9.14] | [10.99] | [11.64] | [8.20] |
| Constant | −2.73*** | −3.84*** | −7.01*** | −14.73*** | −16.17*** | −21.35*** |
| | [−3.76] | [−4.61] | [−8.34] | [−9.40] | [−9.09] | [−7.85] |
| Obs | 15,677 | 18,487 | 10,844 | 15,677 | 18,487 | 10,844 |
| Adj $R^2$ | 0.056 | 0.057 | 0.095 | 0.047 | 0.041 | 0.085 |
| **Panel D: NSOE Dummy and Government Holdings** | | | | | | |
| *NSOE* | 0.63*** | 0.59*** | 0.10 | 1.12*** | 1.33*** | 0.07 |
| | [6.74] | [6.56] | [0.73] | [4.32] | [5.59] | [0.20] |

(*Continued*)

Table VI: Relative performance between non-SOEs and SOEs.

**Table VI**—*Continued*

| Panel D: NSOE Dummy and Government Holdings | | | | | | |
|---|---|---|---|---|---|---|
| *GovtHoldings* | 0.18 | 0.14 | −0.09 | 0.08 | 0.31 | 0.21 |
| | [0.97] | [0.86] | [−0.32] | [0.17] | [0.64] | [0.30] |
| *EquitySize* | 0.18*** | 0.19*** | 0.35*** | 0.78*** | 0.73*** | 1.09*** |
| | [6.02] | [6.17] | [8.61] | [10.99] | [11.08] | [7.80] |
| Constant | −3.60*** | −4.04*** | −7.04*** | −16.27*** | −16.63*** | −21.37*** |
| | [−4.86] | [−4.93] | [−8.50] | [−9.63] | [−9.42] | [−8.00] |
| Obs | 15,677 | 18,487 | 10,844 | 15,677 | 18,487 | 10,844 |
| Adj $R^2$ | 0.065 | 0.064 | 0.095 | 0.051 | 0.046 | 0.085 |

Table VI continued.

## 5.7 Table VII and Table VIII: cross-sectional credit channel and robustness

**Read: Table VII.**

- The postevent performance decline is larger for non-SOEs than SOEs.
- Firms with larger deterioration in the unified default measure suffer larger subsequent performance declines.
- The effect is strongest among firms with high post-2018 credit deterioration.

**Read: Table VIII.**

- The same credit-channel evidence is robust to alternative sample definitions and sorting variables.
- The post-2018Q2 deterioration of non-SOEs relative to SOEs persists across specifications.

**Economic message:** Credit-market stress predicts real firm outcomes. The widening SOE premium is not only a pricing artifact; it is tied to relative real performance.

**Table VII**

**Cross-Sectional Tests on the Credit Channel**

Panel A reports the average postevent performance change in quarterly ROA ($\Delta ROA_{t+\tau}$) for non-SOEs and SOEs separately. "NSOE − SOE" reports the difference in the change in ROA between non-SOEs and SOEs, estimated from regression (18). $\Delta ROA_{t+\tau}$ is computed as the difference between ROA realized $\tau$ quarters after event date $t$= 2018Q2 and its preevent ROA averaged across the four preevent quarters. Panel B reports the predictability regression of the change in quarterly ROA at quarter $t+\tau$ ($\Delta ROA_{t+\tau}$) on the change in default measure at quarter $t$ ($\Delta DM_t$), estimated from regression (19). $\Delta DM_t$ is computed as the difference between the default measure at event time $t$, and its preevent unified default measure averaged across the four preevent quarters. "NSOE − SOE" reports the difference in predictability between non-SOEs and SOEs. Panel C reports the difference in the postevent performance change between non-SOEs and SOEs following in regression (18) for high and low groups separately, sorted by their change in default measure at $t$. "High − Low" reports the difference in the performance gap between the high and low groups, estimated using regression (20). ROA is the ratio of net profit to lagged book assets. *Merton DM* is the inverse of Merton's distance-to-default. *Unified DM* is the inverse of the distance-to-default from our unified model. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

Panel A: Sample Averages of Postevent Performance Change, $\Delta ROA_{t+\tau}$ with $t$ =2018Q2

| | $\tau \in [1,1]$ | $\tau \in [1,2]$ | $\tau \in [1,3]$ | $\tau \in [1,4]$ | $\tau \in [1,5]$ | $\tau \in [1,6]$ | $\tau \in [1,7]$ | $\tau \in [1,8]$ |
|---|---|---|---|---|---|---|---|---|
| NSOE | −0.13*** | −0.81*** | −0.69*** | −0.59*** | −0.56*** | −0.71*** | −0.73*** | −0.67*** |
| | [−2.70] | [−11.74] | [−14.13] | [−14.94] | [−16.18] | [−19.46] | [−22.63] | [−22.45] |
| SOE | −0.03 | −0.20*** | −0.18*** | −0.13*** | −0.14*** | −0.18*** | −0.25*** | −0.22*** |
| | [−0.70] | [−4.24] | [−4.94] | [−4.51] | [−5.47] | [−7.15] | [−10.74] | [−10.28] |
| NSOE − SOE | −0.12* | −0.58*** | −0.49*** | −0.45*** | −0.40*** | −0.51*** | −0.46*** | −0.43*** |
| | [−1.66] | [−6.58] | [−7.68] | [−8.59] | [−8.89] | [−11.18] | [−11.30] | [−11.46] |

*(Continued)*

Table VII: Cross-sectional tests on the credit channel, part 1.

**Table VII**—*Continued*

Panel B: Predicting Postevent Performance Change with $\Delta DM$ in 2018Q2

| | Predictability $\beta^{DM}$ (Unified $\Delta DM$) | | | | Predictability $\beta^{DM}$ (Merton $\Delta DM$) | | | | Predictability $\beta^{DM}$ (Unified − Merton $\Delta DM$) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ |
| NSOE | −5.00*** | −3.51*** | −3.79*** | −3.16*** | −1.34 | −0.55 | −0.48 | −0.28 | −3.53** | −3.39*** | −3.92*** | −3.57*** |
| | [−4.16] | [−5.09] | [−5.94] | [−6.14] | [−1.07] | [−0.81] | [−0.78] | [−0.58] | [−2.09] | [−3.67] | [−4.78] | [−5.35] |
| SOE | −1.86 | −1.07 | −0.99 | −0.52 | −0.52 | −0.09 | 0.04 | 0.09 | −0.14 | −0.52 | −0.72 | −0.54 |
| | [−1.50] | [−1.46] | [−1.58] | [−0.97] | [−0.81] | [−0.22] | [0.10] | [0.32] | [−0.14] | [−0.80] | [−1.30] | [−1.15] |
| NSOE − SOE | −3.32* | −2.48** | −2.95*** | −2.64*** | −0.81 | −0.45 | −0.58 | −0.36 | −3.52* | −2.86*** | −3.11*** | −3.02*** |
| | [−1.91] | [−2.43] | [−3.28] | [−3.53] | [−0.59] | [−0.57] | [−0.83] | [−0.64] | [−1.83] | [−2.61] | [−3.22] | [−3.79] |

Panel C: Difference in Postevent Performance Change between Non-SOEs and SOEs

| | ROA Gap $b^{NSOE}$ (Unified $\Delta DM$) | | | | ROA Gap $b^{NSOE}$ (Merton $\Delta DM$) | | | | ROA Gap $b^{NSOE}$ (Unified − Merton $\Delta DM$) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ |
| High $\Delta DM$ | −0.79*** | −0.61*** | −0.69*** | −0.61*** | −0.73*** | −0.53*** | −0.64*** | −0.54*** | −0.69*** | −0.58*** | −0.62*** | −0.55*** |
| | [−5.84] | [−7.87] | [−10.07] | [−10.81] | [−5.28] | [−6.70] | [−9.07] | [−9.43] | [−5.63] | [−7.75] | [−9.41] | [−10.08] |
| Low $\Delta DM$ | −0.31*** | −0.25*** | −0.30*** | −0.24*** | −0.43*** | −0.38*** | −0.41*** | −0.35*** | −0.42*** | −0.28*** | −0.37*** | −0.28*** |
| | [−2.67] | [−3.46] | [−4.88] | [−4.73] | [−3.95] | [−5.53] | [−6.77] | [−6.93] | [−3.34] | [−3.84] | [−5.68] | [−5.23] |
| High − Low | −0.51*** | −0.38*** | −0.41*** | −0.38*** | −0.22 | −0.09 | −0.14 | −0.12 | −0.44*** | −0.42*** | −0.37*** | −0.38*** |
| | [−3.03] | [−3.85] | [−4.69] | [−5.26] | [−1.32] | [−0.92] | [−1.58] | [−1.59] | [−2.65] | [−4.29] | [−4.17] | [−5.25] |

Table VII: Cross-sectional tests on the credit channel, part 2.

**Table VIII**

**Robustness Tests on Alternative Explanations**

Panel A reports the sensitivity regression of the change in quarterly ROA at quarter $t+\tau$ ($\Delta ROA_{t+\tau}$) on three measures of trade war exposure. "NSOE − SOE" reports the difference in sensitivity between non-SOEs and SOEs. "Performance Gap" columns report the difference in the postevent performance change between non-SOEs and SOEs following regression (18) for high and low groups, sorted by the three measures of trade war exposure. "High − Low" reports the difference between the high and low groups, estimated from regression (20). Panel B reports the sensitivity regression of the change in quarterly ROA at quarter $t+\tau$ ($\Delta ROA_{t+\tau}$) on three deleveraging-motivated sorting variables. The last row reports the difference between the best non-SOEs in the low group against the worst SOEs in the high group. ROA is the ratio of net profit to lagged book assets. Reported in square brackets are $t$-statistics using standard errors clustered by bond and quarter. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

Panel A: U.S.-China Trade War as an Alternative Explanation

| | X = Trade-War-Affected Industries | | | | X = Trade-War CAPM Beta | | | | X = Trade-War Return | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Sensitivity $\beta^X$ | | | | Sensitivity $\beta^X$ | | | | Sensitivity $\beta^X$ | | | |
| | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ | $\tau \in [1,2]$ | $\tau \in [1,4]$ | $\tau \in [1,6]$ | $\tau \in [1,8]$ |
| NSOE | 0.36 | 0.17 | 0.10 | 0.09 | 0.04 | −0.03 | 0.00 | −0.07 | −1.41 | −0.25 | 2.00 | 2.52** |
| | [1.46] | [1.23] | [0.70] | [0.82] | [0.25] | [−0.28] | [−0.01] | [−1.06] | [−0.51] | [−0.15] | [1.30] | [2.03] |
| SOE | 0.04 | 0.20 | 0.10 | 0.13 | −0.21* | −0.11* | −0.09 | −0.06 | −3.53** | 0.91 | 0.43 | −0.16 |
| | [0.16] | [1.06] | [0.66] | [0.95] | [−1.90] | [−1.69] | [−1.63] | [−1.21] | [−2.02] | [0.67] | [0.42] | [−0.19] |
| NSOE − SOE | −0.04 | −0.20* | −0.18** | −0.13* | 0.14 | 0.02 | 0.01 | −0.08 | 3.31 | −0.53 | 2.35 | 3.25** |
| | [−0.26] | [−1.94] | [−2.05] | [−1.77] | [0.77] | [0.21] | [0.07] | [−0.99] | [1.07] | [−0.27] | [1.34] | [2.27] |
| | Performance Gap $b^{NSOE}$ | | | | Performance Gap $b^{NSOE}$ | | | | Performance Gap $b^{NSOE}$ | | | |
| High X | −0.61*** | −0.56*** | −0.59*** | −0.49*** | −0.49*** | −0.41*** | −0.48*** | −0.45*** | −0.61*** | −0.44*** | −0.56*** | −0.51*** |
| | [−4.52] | [−6.90] | [−8.32] | [−8.25] | [−3.99] | [−5.44] | [−7.27] | [−8.13] | [−5.41] | [−6.60] | [−9.29] | [−10.25] |
| Low X | −0.57*** | −0.38*** | −0.45*** | −0.39*** | −0.56*** | −0.41*** | −0.46*** | −0.37*** | −0.51*** | −0.46*** | −0.45*** | −0.35*** |
| | [−4.83] | [−5.43] | [−7.34] | [−7.76] | [−4.67] | [−5.79] | [−7.21] | [−6.91] | [−3.77] | [−5.59] | [−6.41] | [−5.94] |
| High − Low | −0.04 | −0.20* | −0.18** | −0.13* | −0.02 | −0.05 | −0.06 | −0.10 | −0.15 | −0.03 | −0.18** | −0.23*** |
| | [−0.26] | [−1.94] | [−2.05] | [−1.77] | [−0.14] | [−0.55] | [−0.69] | [−1.40] | [−0.85] | [−0.32] | [−1.99] | [−3.09] |

*(Continued)*

Table VIII: Robustness, part 1.

**Table VIII**—*Continued*

Panel B: The 2016 to 2017 SOE Deleveraging as an Alternative Explanation

| | X = Leverage (17Q2–18Q1 avg) | | | | X = $\Delta$Leverage (17Q2–18Q1 − 15Q1–Q4) | | | | X = $\Delta$Leverage (17Q1–Q4 − 15Q4–16Q3) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Sensitivity $\beta^X$ | | | | Sensitivity $\beta^X$ | | | | Sensitivity $\beta^X$ | | | |
| NSOE | 0.19 | 0.28 | 0.19 | 0.32 | −0.27 | −0.40 | −0.22 | −0.20 | −0.64 | −0.69 | −0.54 | −0.61* |
| | [0.39] | [1.02] | [0.77] | [1.60] | [−0.45] | [−1.17] | [−0.73] | [−0.81] | [−0.80] | [−1.54] | [−1.31] | [−1.81] |
| SOE | −0.19 | 0.06 | 0.14 | 0.27* | −0.99* | −0.22 | −0.04 | 0.30 | −1.78** | −0.78 | −0.60 | −0.14 |
| | [−0.60] | [0.32] | [0.83] | [1.83] | [−1.69] | [−0.64] | [−0.13] | [1.18] | [−1.99] | [−1.49] | [−1.26] | [−0.35] |
| NSOE − SOE | 0.36 | 0.22 | 0.13 | 0.08 | 0.59 | −0.27 | −0.34 | −0.60* | 0.87 | −0.10 | −0.22 | −0.68 |
| | [0.69] | [0.72] | [0.48] | [0.38] | [0.71] | [−0.55] | [−0.79] | [−1.73] | [0.74] | [−0.14] | [−0.36] | [−1.34] |
| | Performance Gap $b^{NSOE}$ | | | | Performance Gap $b^{NSOE}$ | | | | Performance Gap $b^{NSOE}$ | | | |
| Low X NSOE | −0.48*** | −0.42*** | −0.50*** | −0.45*** | −0.32** | −0.28*** | −0.36*** | −0.32*** | −0.26* | −0.25*** | −0.36*** | −0.29*** |
| − High X SOE | [−3.58] | [−5.31] | [−7.08] | [−7.75] | [−2.27] | [−3.45] | [−5.05] | [−5.45] | [−1.81] | [−3.00] | [−4.77] | [−4.70] |

Table VIII: Robustness, part 2.

# 6 Short summary

- The paper studies China's corporate bond market to quantify SOEs' funding advantage.
- The SOE premium is the non-SOE minus SOE spread difference after controlling for ratings and characteristics.
- Before 2018, the premium is meaningful but moderate.
- After the April 2018 New Regulations, the premium explodes.
- The regulations reduce shadow-banking support and corporate bond liquidity.
- Non-SOEs are hit harder because they lack bailout protection.
- The model extends He and Xiong rollover risk with a bailout probability.
- Higher liquidity risk makes bailout protection more valuable.
- A unified default measure that incorporates liquidity and bailout explains the SOE premium better than Merton's default measure.
- Credit spreads become more sensitive to credit/default information after 2018Q2.
- SOEs become more sensitive to government support; non-SOEs become more sensitive to credit quality.
- Non-SOE operating performance deteriorates relative to SOEs after the credit shock.
- The paper connects credit-market segmentation to real outcomes.

# 7 Conclusion

## 7.1 Core conclusion

Geng and Pan show that the SOE premium in China's credit market reflects a deep divide between firms with and without perceived government support. The divide becomes much larger after a liquidity shock because bailout protection reduces liquidity-driven default risk for SOEs.

## 7.2 Economic intuition

When debt-market liquidity worsens, firms face higher rollover costs. Non-SOEs have to bear the increased default risk directly. SOEs, however, benefit from the possibility that the government will support repayment. Therefore, the same liquidity shock raises spreads more for non-SOEs than for SOEs.

## 7.3 Takeaway

The paper's main lesson is that implicit government guarantees matter most in bad times. A policy designed to reduce shadow-banking risk can unintentionally deepen credit misallocation by increasing the value of government support and widening the funding gap between SOEs and non-SOEs.